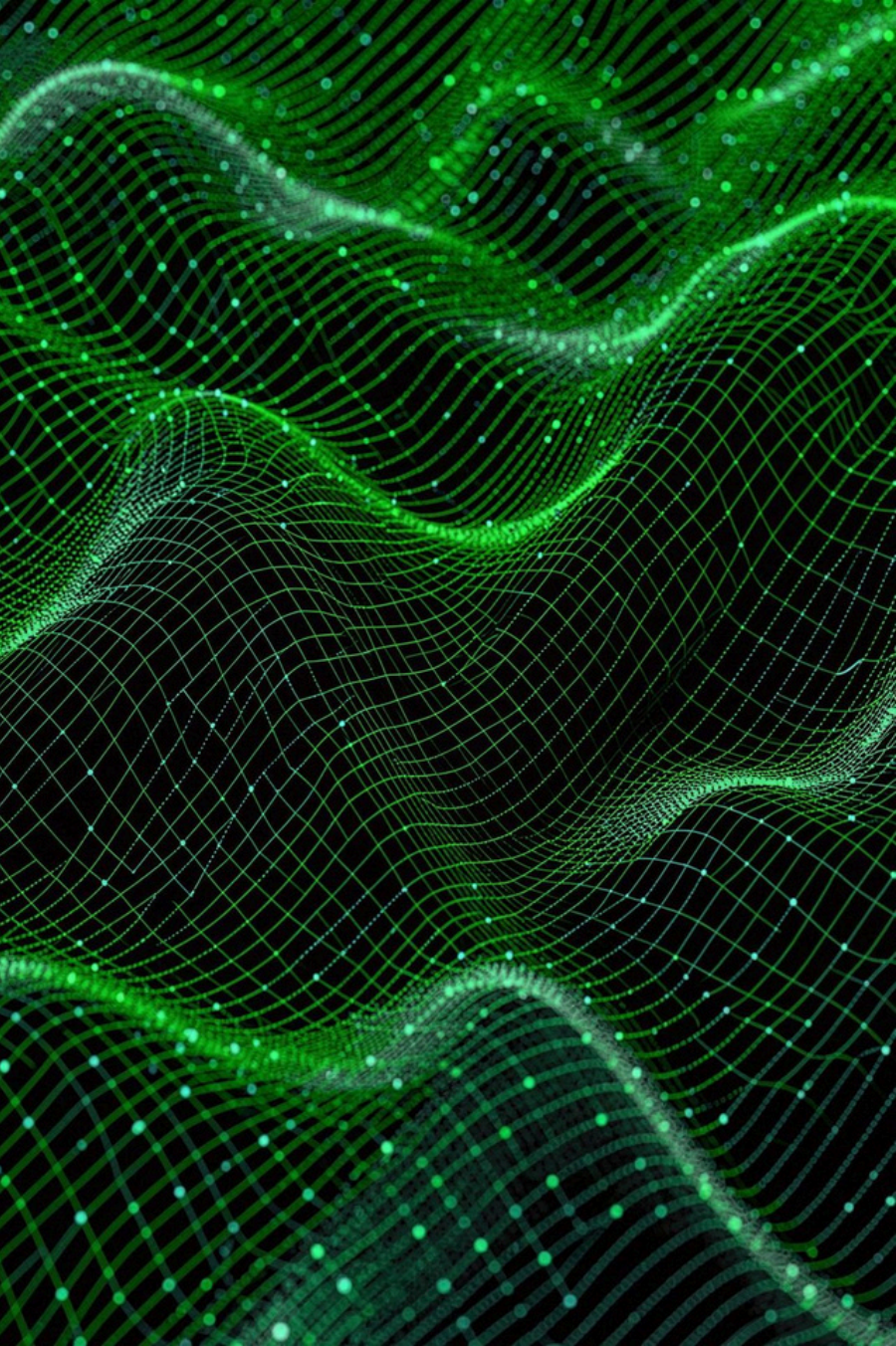


An abstract digital graphic on the left side of the slide, featuring a series of green, wavy lines that resemble a wireframe or a digital landscape. The lines are composed of small dots and connect to form a mesh-like structure. The overall color is a vibrant green, and the pattern has a sense of depth and movement, suggesting a digital or technological theme.

HeatShield-X

AI-Powered Urban Heat Risk, Vulnerability & Intervention Optimization

Team- LOGIC LEGENDS

team member

TOYANSHEE, CHHAVI, CHETNA
YAJVI

Track

Sustainable Future & Smart Cities

"From predicting urban heat to deciding where the city should act first."

Cities Know the Temperature. They Don't Know What to Do Next.

The Problem

Urban heat rising due to dense construction, roads, low vegetation, and accelerating climate change — a systemic urban emergency.

The Gap

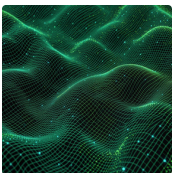
Heat maps show temperature, **not action**. Existing tools stop at diagnosis. No city has a clear answer for *what to do next*.

The Missing Answers

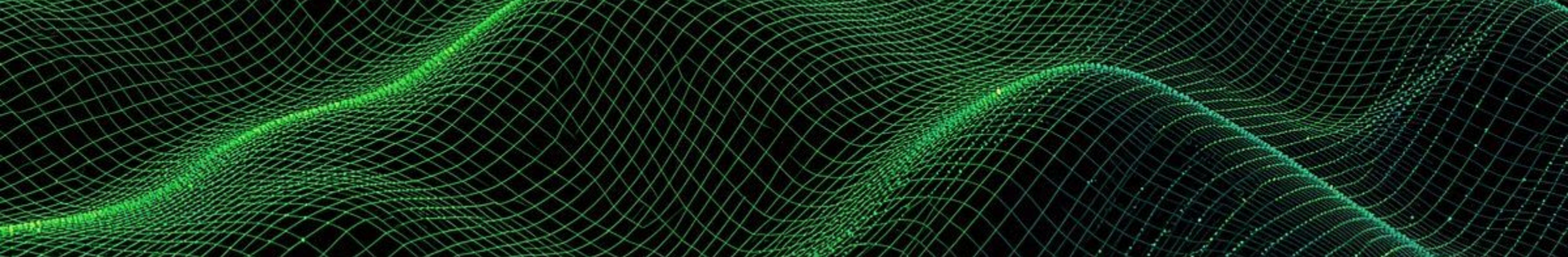
Who is vulnerable? Which area first? What intervention works?
What is feasible within a fixed budget?

The Reality

“We are not claiming to be the first to detect urban heat. We are solving the next problem: deciding what a city can realistically do about it.”



The real challenge is not detecting heat — it is turning heat intelligence into actionable city decisions.



HeatShield-X: The Complete Solution

Satellite Data → Environmental Intelligence → Heat Prediction → Vulnerability Assessment → Intervention Readiness → AI Optimization → Impact Simulation

Core Innovations — What Sets Us Apart



Intervention Readiness

Validates whether recommended actions are actually feasible and implementable in a given zone — before any budget is committed.



Conflict Detection

Rejects unsuitable solutions and suggests alternatives. *E.g., tree plantation rejected → recommend cool roofs + shaded corridors.*



Budget-Aware Optimization

Selects intervention combinations delivering the greatest heat-risk reduction within the city's fixed budget envelope.

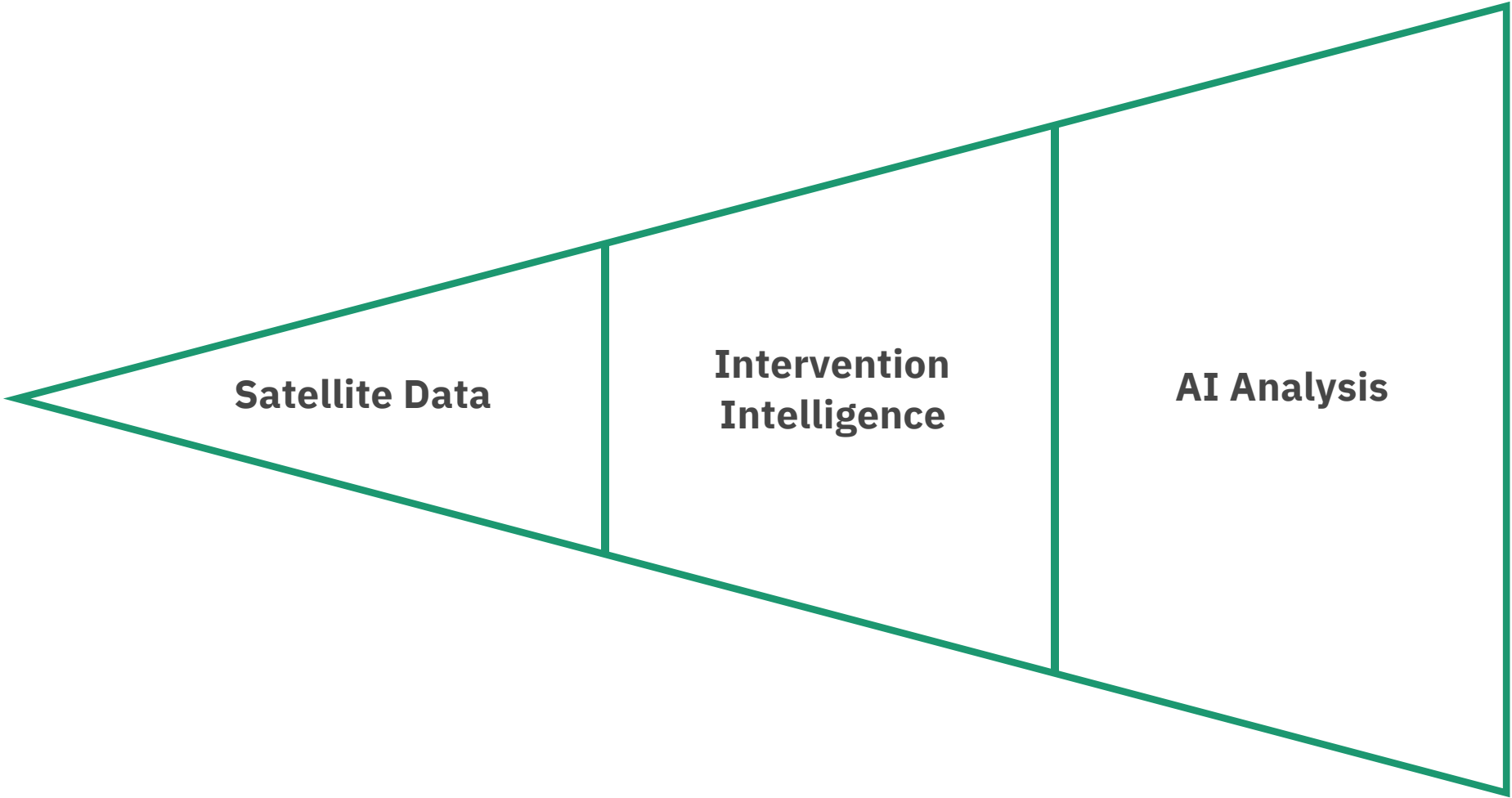


What-If Simulation

Authorities compare predicted outcomes across different intervention scenarios before committing resources.

📄 We don't just identify where it is hot. We identify **where to act, what to do, what NOT to do, who benefits, and how much risk can be reduced.**

How It Works: Five-Layer Architecture



A five-layer flow from raw earth observation to actionable city decisions, using heat-aware analysis and budget-constrained optimization.

01 Satellite Data

Earth observation, thermal imagery, land cover, weather, population indicators

02 AI Analysis

Neural feature extraction, heat-risk prediction, environmental pattern analysis, vulnerability scoring

03 Intervention Intelligence

City planning constraints, space availability, vegetation feasibility, building suitability, water access, infrastructure conflicts

04 Optimization engine

Ranks trees, cool roofs, shade structures, water stations, and reflective surfaces by risk, feasibility, impact, and budget

05 Decision Dashboard

Heat Risk Score, Priority Zone, Recommended Actions, Rejected Interventions, Cost, Predicted Impact, Before vs After simulation

Technology & AI Stack



Data & Geospatial

Satellite imagery · Remote sensing · NDVI · Land Surface Temperature · GIS/geospatial layers



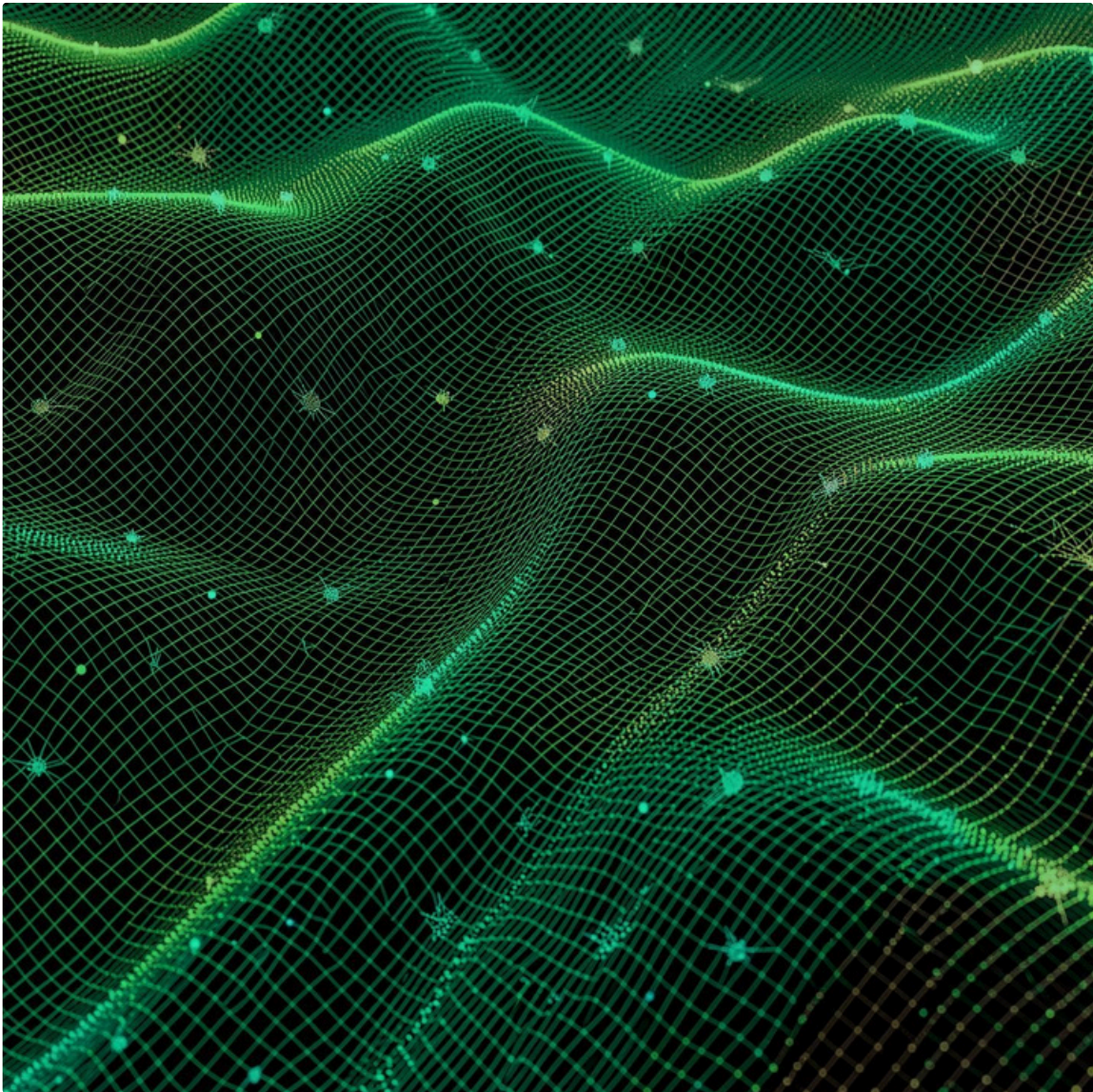
Machine Learning

Heat-risk prediction · Anomaly detection · Vulnerability scoring · Intervention recommendation · Optimization algorithms



Prototype Stack

Python · Pandas · NumPy · Scikit-learn · GeoPandas · Streamlit · Plotly/Folium



Optimization Objective

Maximize

Heat-risk reduction + vulnerable population protected

Subject To

Budget · Land · Water · Infrastructure · Feasibility constraints

Explainable AI

Transparent scoring logic — every zone decision is auditable:

- Low vegetation
- + High density
- + High surface temperature
- + High population exposure
- = **High-Risk Zone**

Prototype & Demo: From Data to Decision

Interactive Heat Risk Map

● Low Risk

● Moderate

● High

● Critical

Zone selection reveals: Heat Risk Score · Vulnerability Level · Intervention Readiness Score

AI Recommendation Engine

Parameter	Value
Trees planted	300
Cool roofs	20
Shaded bus stops	5
Budget	₹10 lakh
Heat Risk Before	87
Heat Risk After	70

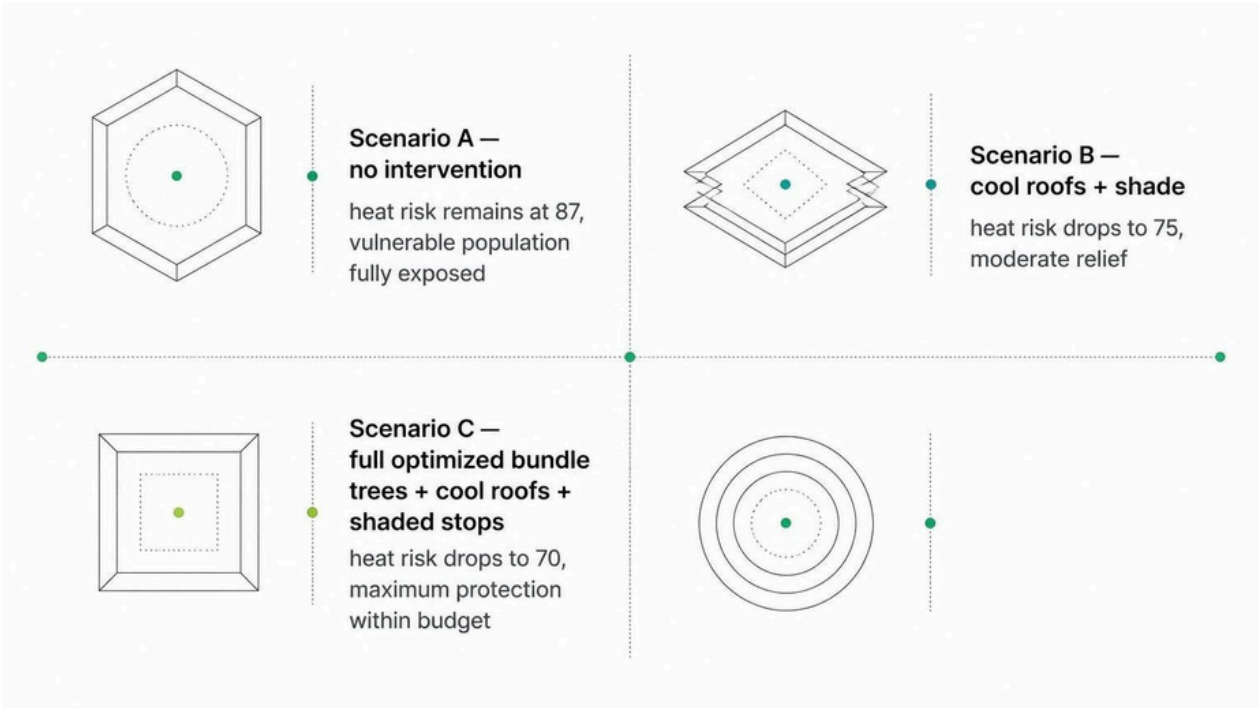
Intelligent Rejection & Alternatives

✗ Rejected: Tree Plantation

Reason: Insufficient space availability + water feasibility score below threshold. System automatically flags and removes this option.

✓ Alternative Recommended

Cool roof installation + shade infrastructure — validated as feasible, costed, and impact-simulated within the same budget envelope.



Prototype uses simulated/demo data, designed for integration with real satellite, weather, and city datasets.

Feasibility & Impact: Scalable Today

Why It's Feasible

Technical Readiness

Built on existing satellite and geospatial technologies. AI models trained from historical and environmental data. Software-first — no specialized hardware required.

Operational Fit

Decision-support platform designed for municipalities. Integrates with existing GIS and city systems. Recommendations update automatically as new data arrives.

Scalability Path

One city → Multiple zones → Multiple cities. Architecture built for horizontal scaling from day one.

Expected Impact



Reduced Heat Exposure

Targeted reduction of heat risk in the highest-vulnerability zones



Vulnerable Community Protection

Better coverage of at-risk populations — elderly, dense settlements, heat islands



Smarter Green Infrastructure

Data-driven planning replaces guesswork in municipal greening programs



Optimized Municipal Budgets

Maximum risk reduction per rupee spent, with full transparency



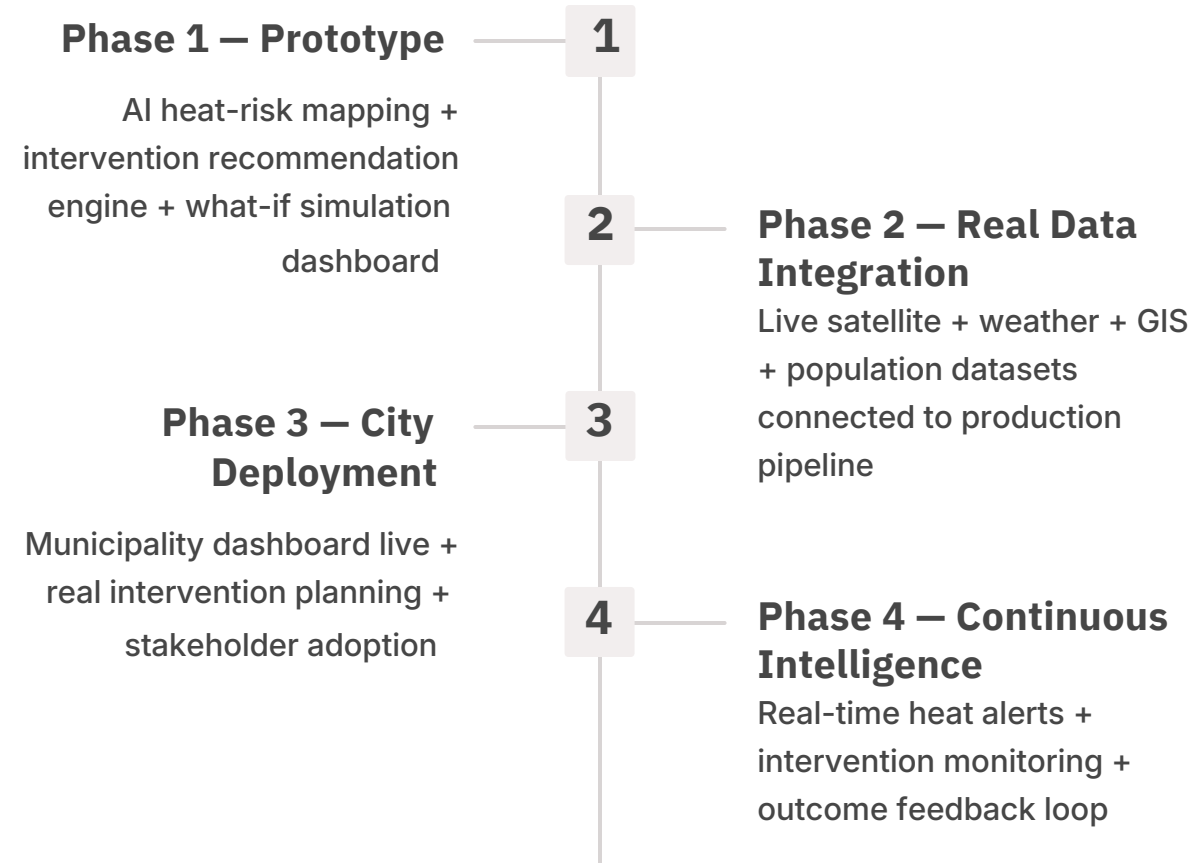
Climate-Resilient Cities

Faster, evidence-based intervention decisions at urban scale



Prototype estimates shown; real-world impact validated through city deployment.

Roadmap & Vision



"A city that doesn't wait for extreme heat to become a crisis."

HeatShield-X turns environmental data into actionable urban resilience.

Predict. Prioritize. Optimize. Protect.

Team

TOYANSHEE CHHAVI CHETNA YAJVI

"A HOTTER CITY NEEDS MORE THAN WARNINGS—IT NEEDS ACTION. TOGETHER, WE CAN PREDICT, PRIORITIZE, AND MAKE THAT ACTION POSSIBLE."

